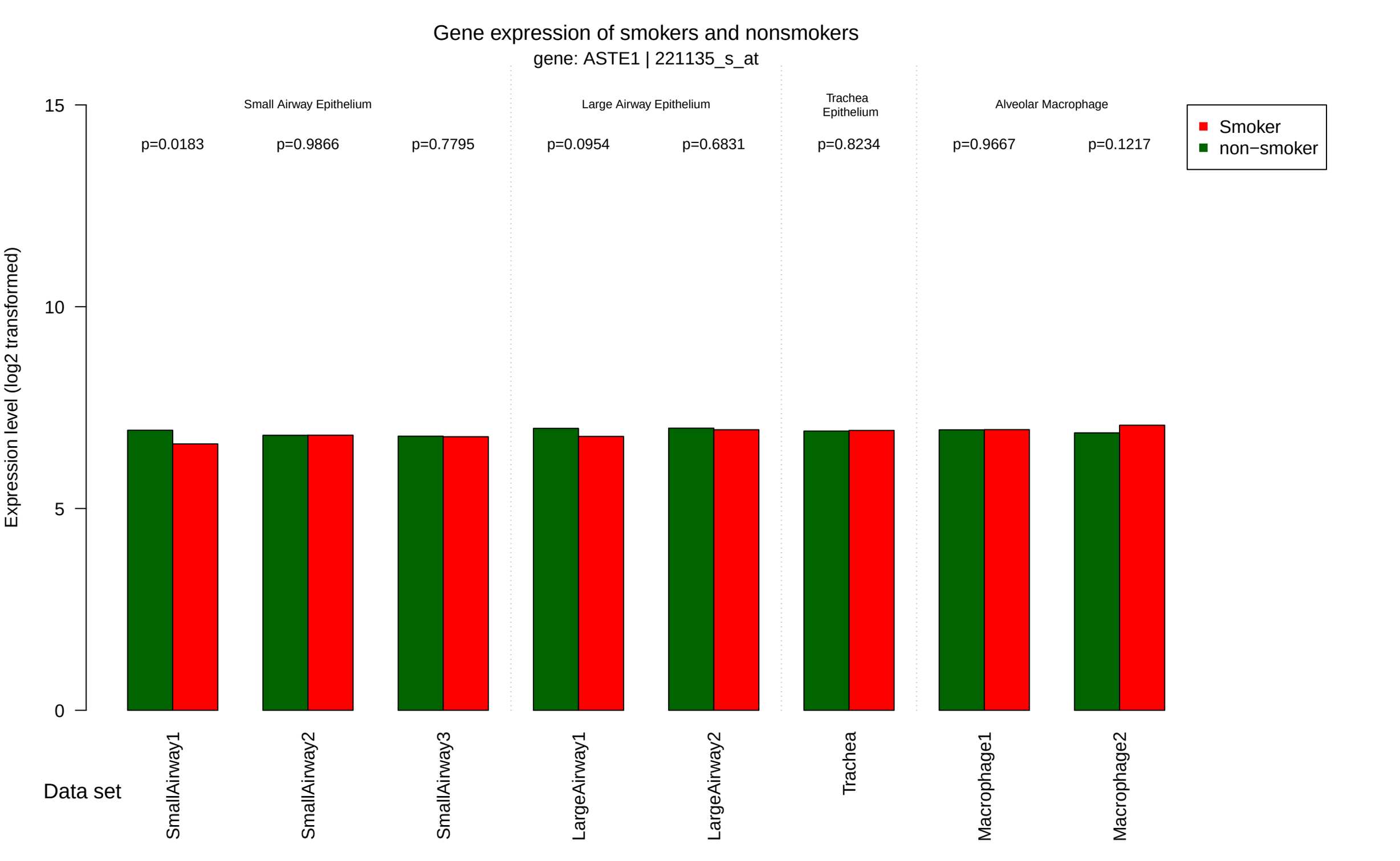
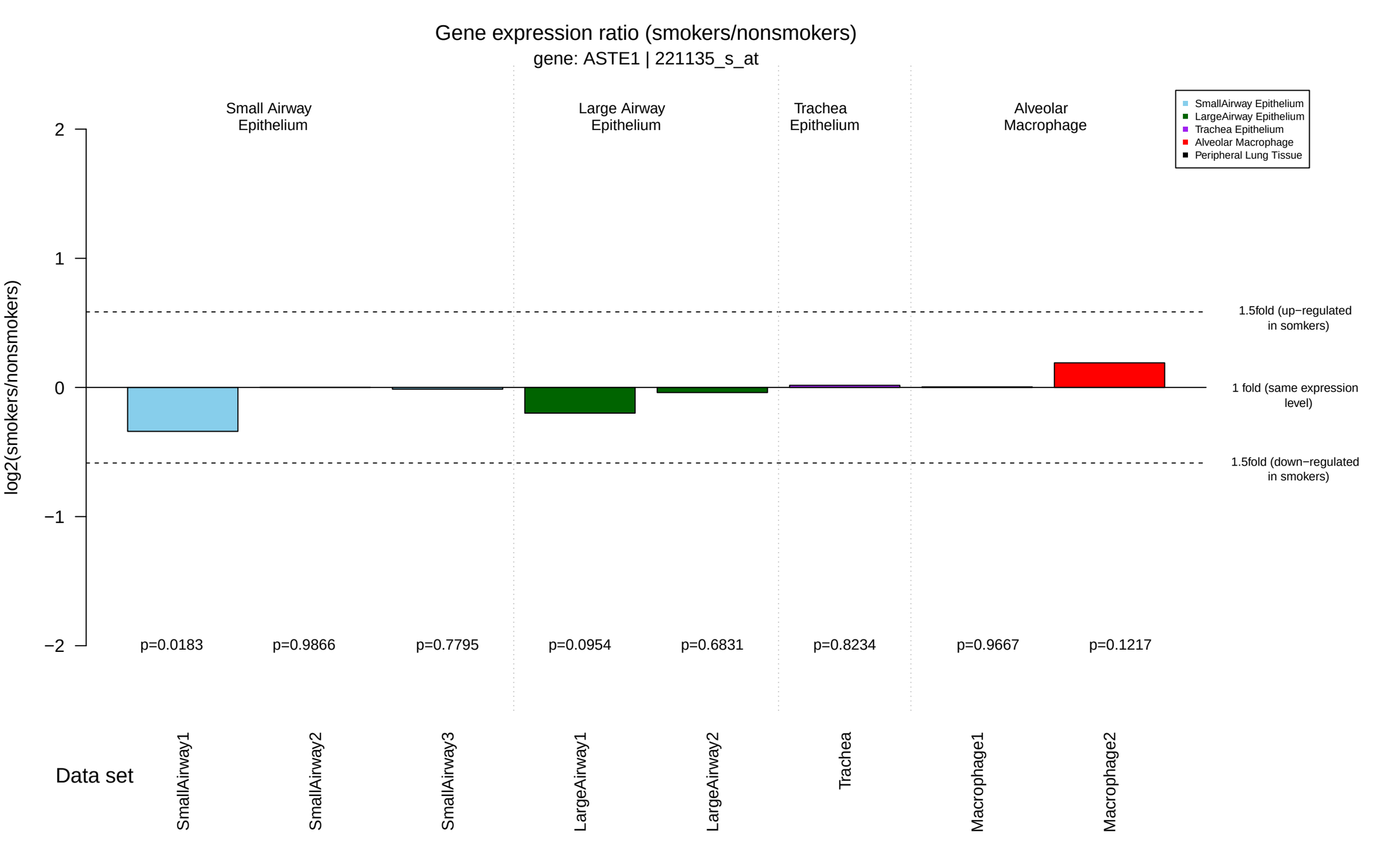
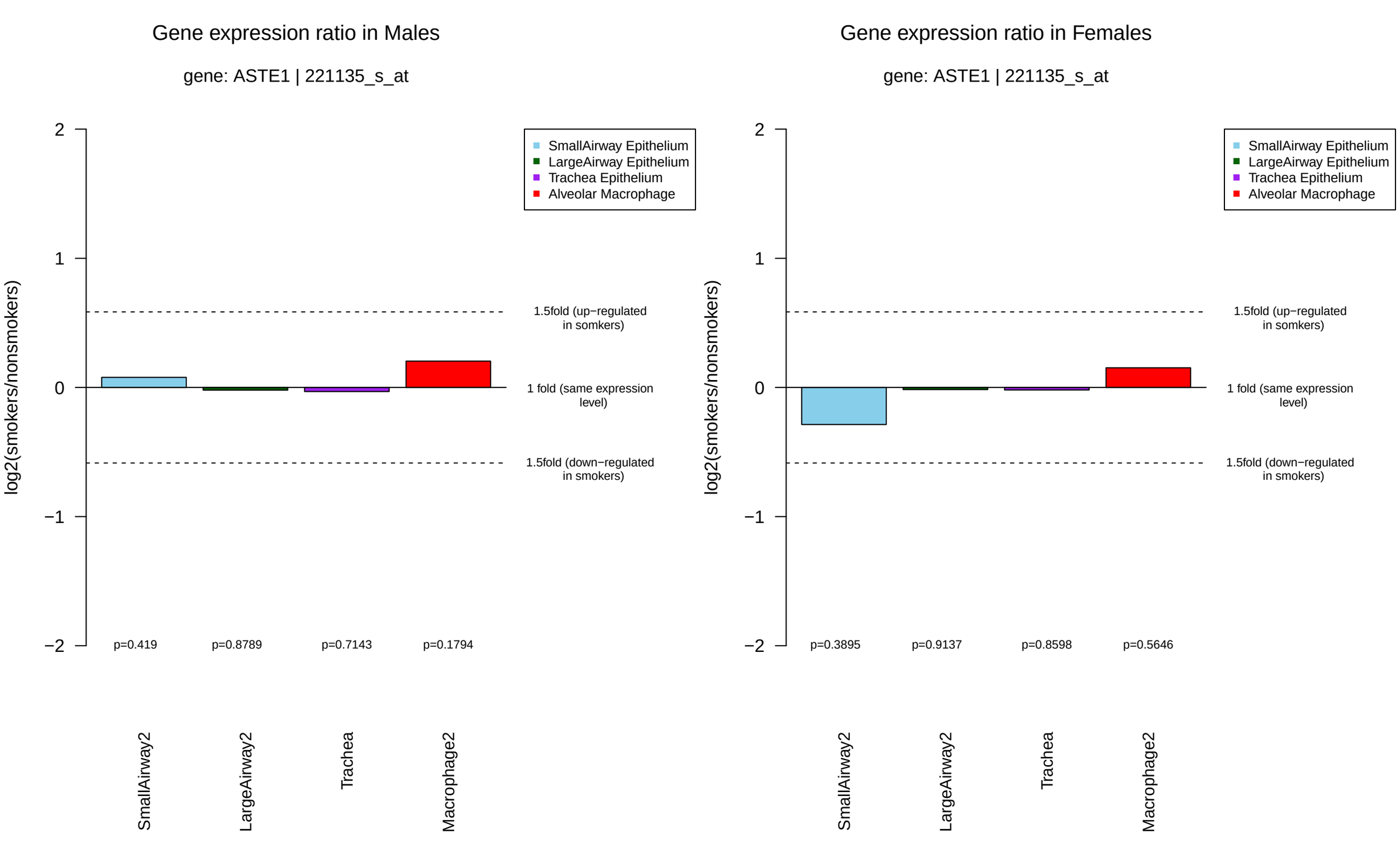
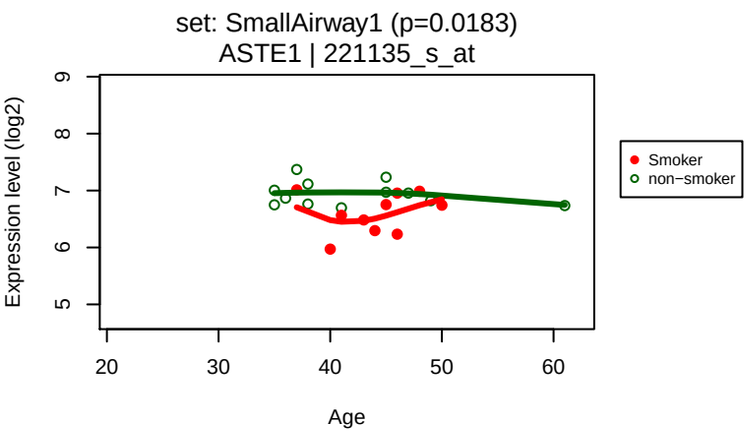
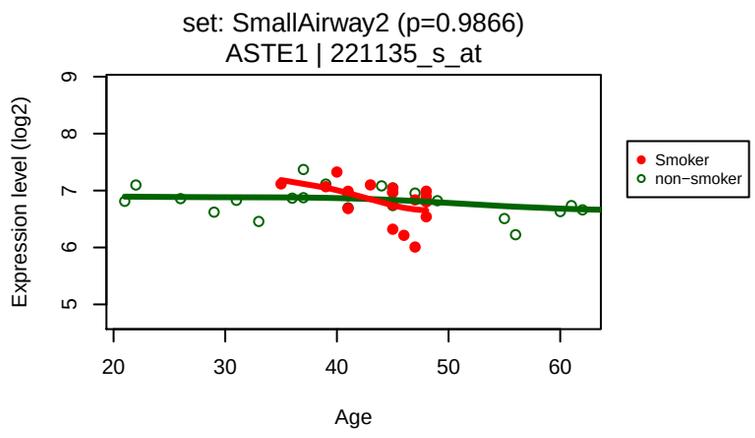
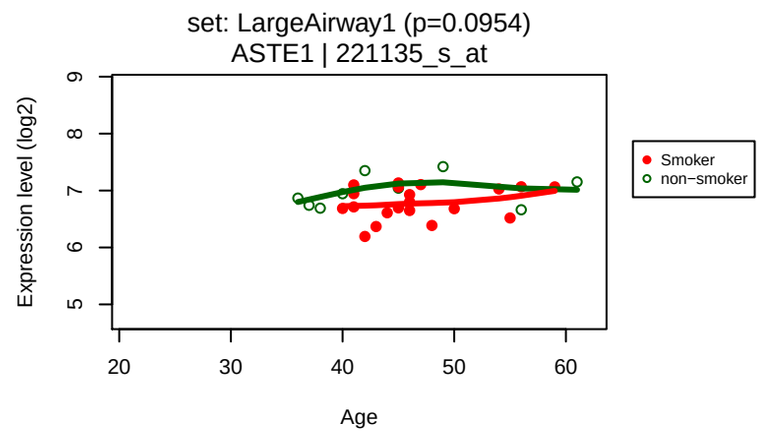

Gene expression of smokers and nonsmokers
gene: ASTE1 | 221135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0183
p=0.9866
p=0.7795
p=0.0954
p=0.6831
p=0.8234
p=0.9667
p=0.1217
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

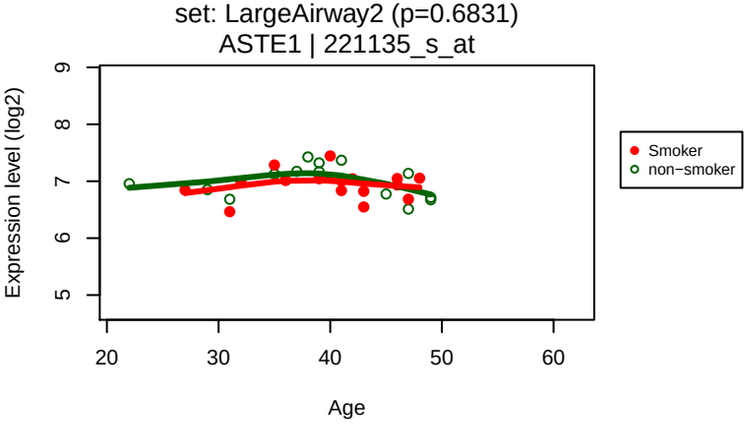

Gene expression ratio (smokers/nonsmokers)
gene: ASTE1 | 221135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0183
p=0.9866
p=0.7795
p=0.0954
p=0.6831
p=0.8234
p=0.9667
p=0.1217
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

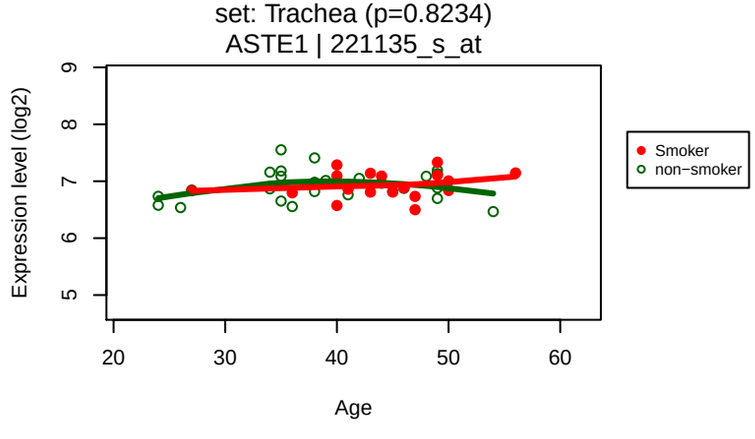

Gene expression ratio in Males
gene: ASTE1 | 221135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.419
p=0.8789
p=0.7143
p=0.1794
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASTE1 | 221135_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3895
p=0.9137
p=0.8598
p=0.5646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

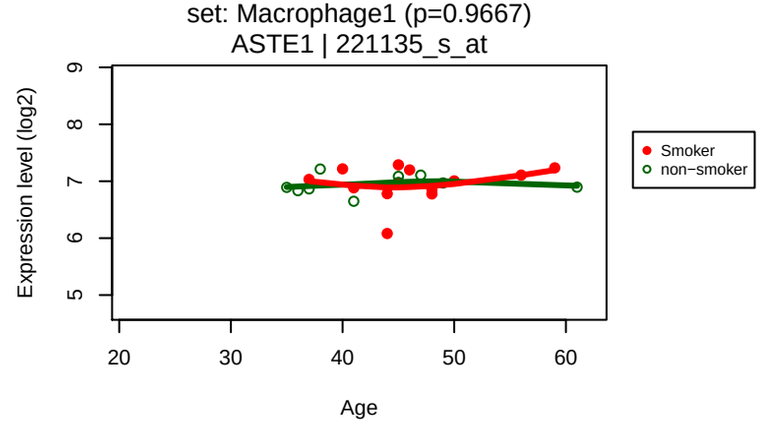

set: SmallAirway1 (p=0.0183)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker

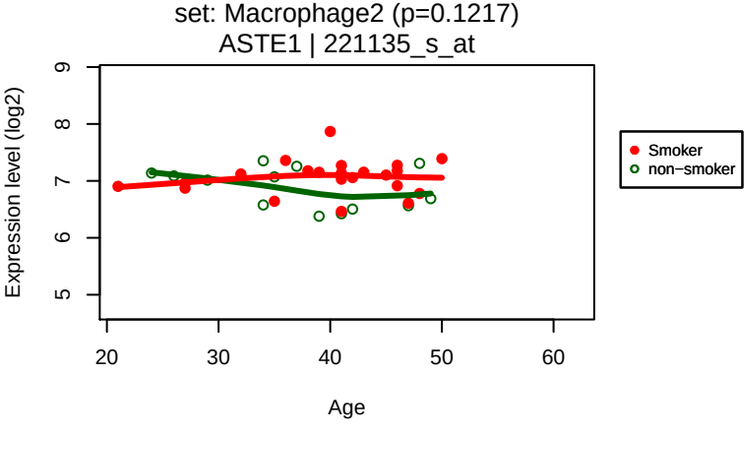

set: SmallAirway2 (p=0.9866)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker

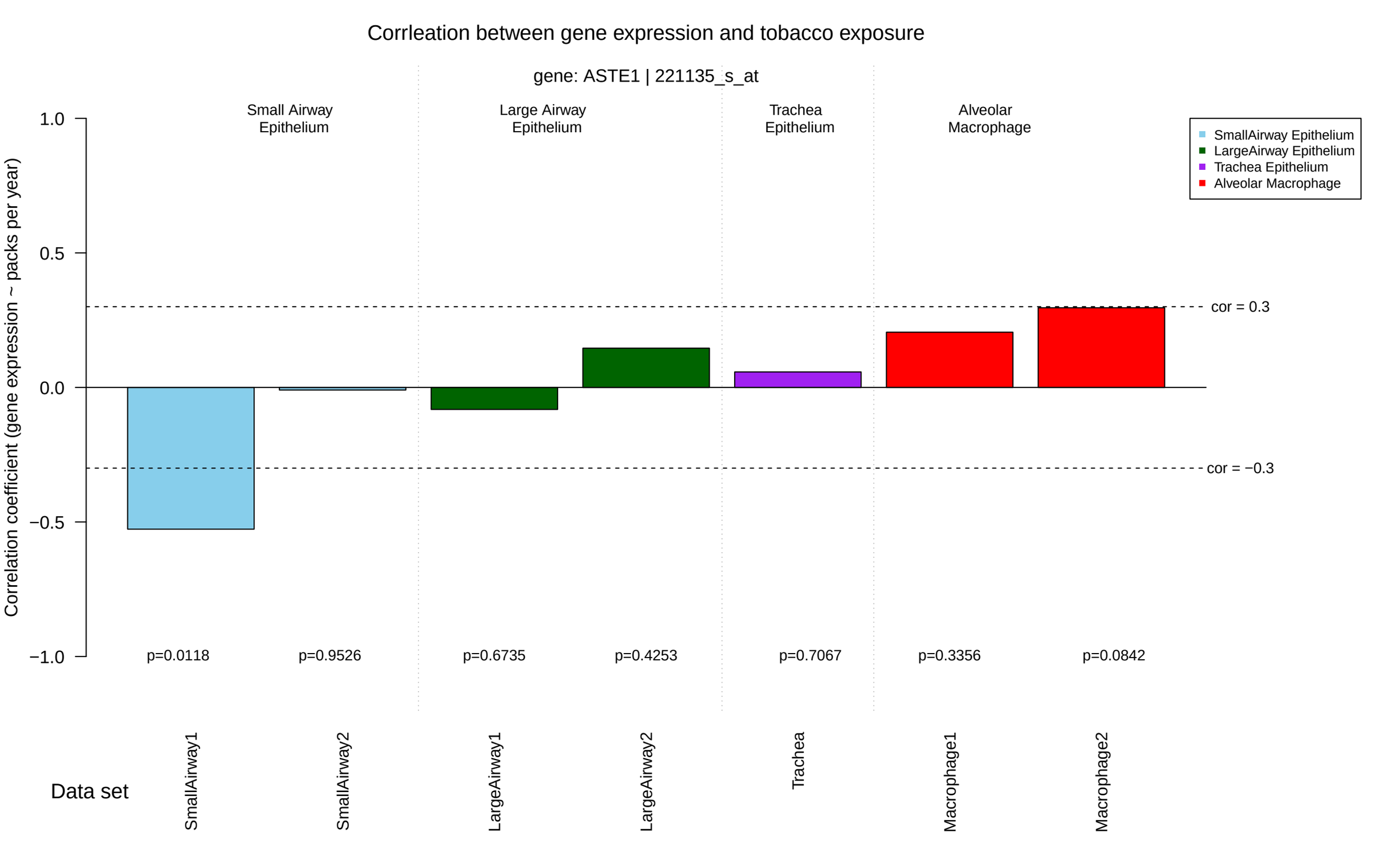

set: LargeAirway1 (p=0.0954)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6831)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8234)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9667)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1217)
ASTE1 | 221135_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASTE1 | 221135_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0118
p=0.9526
p=0.6735
p=0.4253
p=0.7067
p=0.3356
p=0.0842
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2